

# INTRODUCTION TO ALGEBRA AND CALCULUS QUIZ

Revision Guide: Correct Answers and Step-by-Step Solutions

**1. If  $\alpha, \beta$  are zeros of  $x^2 - 7x + 12$ , then  $\alpha + \beta = ?$**

- (a) 12
- (b) 7
- (c) -7
- (d) -12

**Correct answer: (b) 7**

Solution:

- For a quadratic  $ax^2 + bx + c = 0$ , the sum of the zeros is  $\alpha + \beta = -b/a$ .
- Here,  $a = 1$  and  $b = -7$ .
- $\alpha + \beta = -(-7)/1 = 7$ .

**2. Remainder when  $x^3 - 2x^2 + x - 1$  is divided by  $x - 2$ :**

- (a) 1
- (b) 0
- (c) -1
- (d) 5

**Correct answer: (a) 1**

Solution:

- By the Remainder Theorem, divide by  $x - 2$  means substitute  $x = 2$ .
- $f(2) = 2^3 - 2(2^2) + 2 - 1$ .
- $f(2) = 8 - 8 + 2 - 1 = 1$ .
- Therefore, the remainder is 1.

**3. Which of the following is a factor of  $x^3 - 8$ ?**

- (a)  $x + 2$
- (b)  $x - 2$
- (c)  $x^2 + 2x + 4$
- (d) Both B and C

**Correct answer: (d) Both B and C**

Solution:

- $x^3 - 8 = x^3 - 2^3$ .
- Use  $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ .
- $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$ .
- So both  $x - 2$  and  $x^2 + 2x + 4$  are factors.

**4. If  $(x - 1)$  is a factor of  $x^3 - 2x^2 + kx - 4$ , then  $k = ?$**

- (a) 1
- (b) 2
- (c) 5
- (d) 7

**Correct answer: (c) 5**

Solution:

- If  $x - 1$  is a factor, then  $f(1) = 0$ .
- $f(1) = 1^3 - 2(1^2) + k(1) - 4 = 0$ .
- $1 - 2 + k - 4 = 0$ .
- $k - 5 = 0$ , so  $k = 5$ .

**5.  $\lim_{x \rightarrow 2} [(3x - 5)] = ?$**

- (a) 0
- (b) 2
- (c) 1
- (d) -1

**Correct answer: (c) 1**

Solution:

- The expression  $3x - 5$  is continuous, so substitute  $x = 2$ .
- $3(2) - 5 = 6 - 5 = 1$ .
- Therefore, the limit is 1.

**6. The first derivative of  $x$  is:**

- (a)  $1/(2x)$
- (b)  $1/x$
- (c)  $2/x$
- (d)  $x/2$

**Correct answer: (a)  $1/(2x)$**

Solution:

- Rewrite  $x$  as  $x^{(1/2)}$ .
- Using  $d/dx(x^n) = nx^{(n-1)}$ ,  $d/dx(x^{(1/2)}) = (1/2)x^{(-1/2)}$ .
- $x^{(-1/2)} = 1/x$ .
- Therefore,  $d/dx(x) = 1/(2x)$ .

**7.  $\lim_{x \rightarrow 3} [(x^2 - 9)/(x - 3)] = ?$**

- (a) 0
- (b) 6
- (c) 3
- (d) does not exist

**Correct answer: (b) 6**

Solution:

- Factor the numerator:  $x^2 - 9 = (x - 3)(x + 3)$ .
- $(x^2 - 9)/(x - 3) = [(x - 3)(x + 3)]/(x - 3)$ .
- Cancel  $x - 3$  to get  $x + 3$ .
- Substitute  $x = 3$ :  $3 + 3 = 6$ .

**8.  $\frac{d}{dx}(2x^5 - 3x^3 + x)$  at  $x = 1$  is:**

- (a) 2
- (b) 5
- (c) 7
- (d) 9

**Correct answer: (a) 2**

Solution:

- Differentiate term by term.
- $\frac{d}{dx}(2x^5 - 3x^3 + x) = 10x^4 - 9x^2 + 1$ .
- At  $x = 1$ :  $10(1^4) - 9(1^2) + 1 = 10 - 9 + 1 = 2$ .

**9. If  $f(x) = 3x^2 - 2x + 5$ , find  $f'(x)$ .**

- (a)  $6x - 2$
- (b)  $3x - 2$
- (c)  $6x + 2$
- (d)  $3x + 2$

**Correct answer: (a)  $6x - 2$**

Solution:

- Differentiate each term.
- $\frac{d}{dx}(3x^2) = 6x$ .
- $\frac{d}{dx}(-2x) = -2$  and  $\frac{d}{dx}(5) = 0$ .
- Therefore,  $f'(x) = 6x - 2$ .

**10. If  $y = (3x + 2)^2$ , then  $\frac{dy}{dx} =$**

- (a)  $9x + 6$
- (b)  $6x + 4$
- (c)  $18x + 12$
- (d)  $9x^2 + 12x + 4$

**Correct answer: (c)  $18x + 12$**

Solution:

- Let  $u = 3x + 2$ , so  $y = u^2$ .
- $\frac{dy}{du} = 2u$  and  $\frac{du}{dx} = 3$ .
- $\frac{dy}{dx} = 2(3x + 2)(3) = 6(3x + 2)$ .
- $\frac{dy}{dx} = 18x + 12$ .

**11. Find the integral of  $f(x) = 2x + 3$  with respect to  $x$ .**

- (a)  $x^2 - 3x + C$
- (b)  $2x^2 - 3x + C$
- (c)  $x^2 + 3x + C$
- (d)  $3x^3 + 5x + C$

**Correct answer: (c)  $x^2 + 3x + C$**

Solution:

- Integrate term by term:  $(2x + 3) dx = 2x dx + 3 dx$ .
- $2x dx = x^2$ .
- $3 dx = 3x$ .
- Therefore,  $(2x + 3) dx = x^2 + 3x + C$ .

**12. Integrate the function  $f(x) = x + 1/x^2$ .**

- (a)  $x^2/2 - 1/x + C$
- (b)  $x^2/2 + 1/x + C$
- (c)  $x^2 - 1/x + C$
- (d)  $x^2 + 1/x + C$

**Correct answer: (a)  $x^2/2 - 1/x + C$**

Solution:

- Rewrite  $1/x^2$  as  $x^{-2}$ .
- $(x + x^{-2}) dx = x dx + x^{-2} dx$ .
- $x dx = x^2/2$ .
- $x^{-2} dx = x^{-1}/(-1) = -x^{-1} = -1/x$ .
- Therefore, the integral is  $x^2/2 - 1/x + C$ .

**13. Degree of  $7x^5 - 2x^3 + x - 9$  is:**

- (a) 3
- (b) 4
- (c) 5
- (d) 1

**Correct answer: (c) 5**

Solution:

- The degree of a polynomial is the highest power of  $x$ .
- In  $7x^5 - 2x^3 + x - 9$ , the highest power is 5.
- Therefore, the degree is 5.

**14. Which is a polynomial?**

- (a)  $x^2 + x + 1$
- (b)  $2x - 1/x$
- (c)  $3x^4 - 2x + 5$
- (d)  $|x| + 1$

**Correct answer: (c)  $3x^4 - 2x + 5$**

Solution:

- A polynomial has variables with non-negative whole-number powers only.
- $x = x^{1/2}$ , so option (a) is not a polynomial.
- $1/x = x^{-1}$ , so option (b) is not a polynomial.
- $|x|$  is not written as a polynomial expression, so option (d) is not a polynomial.
- Option (c) has powers 4, 1, and 0, so it is a polynomial.

**15. Solve:  $5 \leq 2x - 1 \leq 9$**

- (a)  $[2, 5]$
- (b)  $(2, 5)$
- (c)  $[3, 5]$
- (d)  $(3, 5]$

**Correct answer: (c)  $[3, 5]$**

Solution:

- Start with  $5 \leq 2x - 1 \leq 9$ .
- Add 1 to all sides:  $6 \leq 2x \leq 10$ .
- Divide all sides by 2:  $3 \leq x \leq 5$ .
- In interval form, the solution is  $[3, 5]$ .

**16. For what value of k does the system have infinite solutions?  $2x + 3y = 5$  and  $4x + 6y = k$**

- (a) 5
- (b) 10
- (c) 15
- (d) 20

**Correct answer: (b) 10**

Solution:

- For infinite solutions, the second equation must be exactly the same line as the first.
- Multiply  $2x + 3y = 5$  by 2.
- This gives  $4x + 6y = 10$ .
- Therefore,  $k = 10$ .

**17. Solve:  $|x - 3| = 5$**

- (a)  $x = 8$
- (b)  $x = -2$
- (c)  $x = 8$  or  $x = -2$
- (d)  $x = 3$

**Correct answer: (c)  $x = 8$  or  $x = -2$**

Solution:

- If  $|x - 3| = 5$ , then  $x - 3 = 5$  or  $x - 3 = -5$ .
- From  $x - 3 = 5$ ,  $x = 8$ .
- From  $x - 3 = -5$ ,  $x = -2$ .
- Therefore,  $x = 8$  or  $x = -2$ .

**18. Which line is parallel to  $y = -3x + 4$ ?**

- (a)  $y = 3x + 1$
- (b)  $y = -1/3x + 2$
- (c)  $y = -3x + 7$
- (d)  $y = x - 3$

**Correct answer: (c)  $y = -3x + 7$**

Solution:

- Parallel lines have the same slope.
- The slope of  $y = -3x + 4$  is  $-3$ .
- Option (c),  $y = -3x + 7$ , also has slope  $-3$ .
- Therefore, option (c) is parallel to the given line.

**19. Slope or gradient of line  $2x - 5y = 10$  is**

- (a)  $2/5$
- (b)  $-2/5$
- (c)  $5/2$
- (d)  $-5/2$

**Correct answer: (a)  $2/5$**

Solution:

- Write the equation in the form  $y = mx + c$ .
- $2x - 5y = 10$ .
- $-5y = -2x + 10$ .
- Divide by  $-5$ :  $y = (2/5)x - 2$ .
- The gradient is the coefficient of  $x$ , so the gradient is  $2/5$ .

**20. Solve the system:  $x + 2y = 8$  and  $3x - y = 3$**

- (a) (2, 3)
- (b) (3, 2)
- (c) (1, 3.5)
- (d) (4, 2)

**Correct answer: (a) (2, 3)**

Solution:

- From  $3x - y = 3$ , make  $y$  the subject:  $y = 3x - 3$ .
- Substitute into  $x + 2y = 8$ .
- $x + 2(3x - 3) = 8$ .
- $x + 6x - 6 = 8$ , so  $7x = 14$ .
- $x = 2$ . Then  $y = 3(2) - 3 = 3$ .
- Therefore, the solution is (2, 3).

**21. Solve inequality:  $3x - 5 < 4$**

- (a)  $x < 3$
- (b)  $x > 3$
- (c)  $x \geq 3$
- (d)  $x \leq 3$

**Correct answer: (a)  $x < 3$**

Solution:

- Start with  $3x - 5 < 4$ .
- Add 5 to both sides:  $3x < 9$ .
- Divide by 3:  $x < 3$ .

**22. Which of the following is an irrational number?**

- (a)  $22/7$
- (b) 9
- (c) 5
- (d)  $27$

**Correct answer: (d) 27**

Solution:

- $22/7$  is a fraction, so it is rational.
- $9 = 3$ , so it is rational.
- 5 is an integer, so it is rational.
- $27 = (9 \times 3) = 33$ . Since 3 is irrational, 27 is irrational.

**23. The decimal expansion of  $1/8$  is:**

- (a) Terminating
- (b) Non-terminating repeating
- (c) Non-terminating non-repeating
- (d) Infinite

**Correct answer: (a) Terminating**

Solution:

- Calculate  $1 \div 8$ .
- $1/8 = 0.125$ .
- Because 0.125 ends, the decimal expansion is terminating.

**24. For any real numbers  $a$ ,  $b$ ,  $c$ , the property  $(a + b) + c = a + (b + c)$  is called**

- (a) Commutative
- (b) Associative
- (c) Distributive
- (d) Identity

**Correct answer: (b) Associative**

Solution:

- The grouping changes from  $(a + b) + c$  to  $a + (b + c)$ .
- The order of the terms does not change.
- A change in grouping is called the associative property.

**25. The set of all integers is a subset of:**

- (a) Natural numbers
- (b) Irrational numbers
- (c) Rational numbers
- (d) Whole numbers

**Correct answer: (c) Rational numbers**

Solution:

- Every integer can be written as a fraction with denominator 1.
- For example,  $-3 = -3/1$ ,  $0 = 0/1$ , and  $5 = 5/1$ .
- Therefore, all integers are rational numbers.

**26. The density property of real numbers states**

- (a) Between any two integers there is another integer
- (b) Between any two distinct real numbers there is another real number
- (c) Real numbers are countable
- (d) Real numbers are only rational

**Correct answer: (b) Between any two distinct real numbers there is another real number**

Solution:

- The density property means real numbers are packed closely on the number line.
- Between any two different real numbers, another real number can be found.
- Therefore, option (b) is correct.

**27. If  $x$  is a real number, then  $x^2 \geq 0$  is:**

- (a) Always false
- (b) Sometimes true
- (c) Always true
- (d) Only true for positive  $x$

**Correct answer: (c) Always true**

Solution:

- The square of a positive number is positive.
- The square of a negative number is also positive.
- The square of 0 is 0.
- Therefore, for every real number  $x$ ,  $x^2 \geq 0$  is always true.

**28. The additive inverse of  $-5/3$  is**

- (a)  $-3/5$
- (b)  $3/5$
- (c)  $5/3$
- (d)  $-5/3$

**Correct answer: (c)  $5/3$**

Solution:

- The additive inverse of a number is the number that adds with it to give 0.
- The opposite of  $-5/3$  is  $5/3$ .
- $-5/3 + 5/3 = 0$ .
- Therefore, the additive inverse is  $5/3$ .

**29. Which set is closed under addition?**

- (a) Odd integers
- (b) Irrational numbers
- (c) Even integers
- (d) Prime numbers

**Correct answer: (c) Even integers**

Solution:

- A set is closed under addition if adding any two members of the set gives another member of the same set.
- Even integer + even integer = even integer.
- Odd integers are not closed because odd + odd = even.
- Prime numbers are not closed because  $2 + 3 = 5$ , but  $3 + 5 = 8$ , and 8 is not prime.
- Therefore, even integers are closed under addition.

**30. Solve:  $5x - 9 = 3x + 7$**

- (a)  $x = 4$
- (b)  $x = 8$
- (c)  $x = 2$
- (d)  $x = 6$

**Correct answer: (b)  $x = 8$**

Solution:

- Start with  $5x - 9 = 3x + 7$ .
- Subtract  $3x$  from both sides:  $2x - 9 = 7$ .
- Add 9 to both sides:  $2x = 16$ .
- Divide by 2:  $x = 8$ .